Cortico-Cerebellar Reservoir Project

Master Compendium and Open-Science Repository Workflow

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Abstract

This master document compiles the final theoretical and computational architecture of the **Cortico-Cerebellar Reservoir Project**. Following an exhaustive autonomous iterative search across the mathematical landscape, the project successfully isolated the **Unified Temporal Integration** model, which definitively outperforms the Win-Stay/Lose-Shift (WSLS) behavioral heuristic ($p = 0.00066$) via synaptic eligibility traces and deep cerebellar leaky integration. This volume serves as the comprehensive open-science reference, detailing the repository architecture, data integrity protocols, and version control workflows.

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1 Repository Architecture & Version Control Workflow

To ensure complete scientific reproducibility, transparency, and computational integrity, the codebase is structured according to strict open-science standards. Following the discovery of the Unified Temporal Integration model, the repository was meticulously cleaned to isolate the optimal implementation.

1.1 1. Folder Ontology

The repository enforces a modular, separation-of-concerns directory tree:

- `data/`: Dedicated strictly to empirical and structured datasets.
  - `data/raw/`: **IMMUTABLE RAW DATA.** Contains the raw empirical behavioral logs (`behavioral_compilate.csv`). Files in this directory must **never** be edited, overwritten, or programmatically modified.
- `src/`: Executable source code implementing mathematical and biophysical algorithms.
  - `src/R/`: High-level orchestration scripts. It contains `run_unified_temporal_integration_cv.R`, the final optimized implementation of the Unified Temporal Integration model using C++ Rcpp and CMA-ES cross-validation.
- `results/`: Artifacts and statistical outputs generated by the execution pipelines.
  - `results/tables/`: CSV statistical ledgers, fixed-effects summaries, and cross-validation performance benchmarks (e.g., `iteration_6_n30.csv`).
- `docs/`: Formal academic manuscripts, LaTeX source code, and compiled theoretical monographs. Contains this master compilation document (`read_output.pdf`) and the mathematical formalization of the final model.

1.2 2. Git Version Control & Collaboration Protocols

To maintain a clean, traceable, and reversible commit history on GitHub, contributors must adhere to the following version control protocols:

1. **Atomic, Logical Commits:** Every commit must represent a single, self-contained functional change. Use standardized conventional commit prefixes:
   - `feat:`: Introduction of new modeling features or statistical decoders (e.g., `feat(models): add unified temporal integration script`).
   - `fix:`: Correction of numerical, mathematical, or algorithmic edge cases.
   - `docs:`: Updates to manuscripts, LaTeX reports, and markdown documentation.
   - `chore:`: Structural maintenance, directory reorganization, or dependency updates (e.g., `chore(cleanup): remove deprecated modeling scripts`).
2. **Branching Strategy:** Direct commits to the `main` branch are strictly prohibited for experimental modifications. All developments must occur on descriptive topic branches:
   - `feature/unified-temporal-integration`

   Branches are merged into `main` only after passing full test suites and code reviews.
3. **Synchronization Protocol (Pull Before Push):** Before pushing local commits, developers must synchronize with the remote repository using rebase to prevent unnecessary merge bubbles:

```
git checkout feature/branch-name
git fetch origin
git rebase origin/main
git push origin feature/branch-name
```

Unified Temporal Integration in Cortico-Cerebellar Kinematics

Mathematical Formalization of the Optimal Stochastic Model

Theoretical Neuro-Computation & Dynamical Systems Group

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Abstract

This document formally defines the final unified architecture for the Evidential Cortico-Cerebellar Model (ECCM), termed *Unified Temporal Integration*. Discovered autonomously via a 6-stage iterative mathematical search, this formulation solves the catastrophic variance problems inherent in linear kinematic readout mappings. By combining presynaptic eligibility traces for temporal credit assignment with postsynaptic deep cerebellar nuclei (DCN) leaky integration, the model achieves superior out-of-sample Negative Log-Likelihood (NLL) and strictly significant superiority over the Win-Stay/Lose-Shift (WSLS) behavioral heuristic ($p = 0.00066$).

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1 1. The Failure of Instantaneous Kinematics

Early iterations of the ECCM mapped the instantaneous population difference of the Purkinje cell layers directly to the drift rate v_t of a Drift-Diffusion Model (DDM). This led to extreme trial-to-trial variance: aggressive shifts caused explosive linear drift rates, resulting in statistically insignificant model fits. Mathematical bounds (e.g., hyperbolic tangent functions) artificially suppressed the model's confidence. The biological solution lies in temporal smoothing at two distinct anatomical loci: the synapse and the readout.

2 2. Formal Specification of Unified Temporal Integration

2.1 2.1. Anti-Correlated Climbing Fiber Error

The climbing fiber error signal ΔIO is mapped zero-sum across the two choice representations. A loss for Choice 1 is treated symmetrically as a win for Choice 2:

$$\Delta IO_1 = \text{target}_1 - y_t^{PC1} \tag{1}$$

$$\Delta IO_2 = \text{target}_2 - y_t^{PC2} \tag{2}$$

where $\text{target}_1 = -\text{target}_2$ under anti-correlated mappings.

2.2 2.2. Presynaptic Eligibility Traces ($e_{k,t}$)

To bridge the temporal gap across the inter-trial interval, Molecular Layer Interneuron (MLI) activity $h_{k,t}^{MLI}$ leaves a synaptic eligibility trace. The trace decays by γ_e over time:

$$e_{k,t} = \gamma_e \cdot e_{k,t-1} + h_{k,t}^{MLI} \tag{3}$$

Synaptic plasticity at the parallel fiber to Purkinje cell synapse updates via the trace:

$$\Delta W_{k,t}^{PF1} = \eta \cdot \Delta IO_1 \cdot e_{k,t} - \lambda W_{k,t}^{PF1} \tag{4}$$
$$\Delta W_{k,t}^{PF2} = \eta \cdot \Delta IO_2 \cdot e_{k,t} - \lambda W_{k,t}^{PF2} \tag{5}$$

2.3 2.3. Postsynaptic Deep Cerebellar Nuclei (DCN) Integration

The instantaneous Purkinje cell difference $y_t^{PC1} - y_t^{PC2}$ is not passed directly to the motor cortex. Instead, it is integrated by the DCN, acting as a low-pass filter:

$$D_t = (1 - \alpha_{DCN})D_{t-1} + \alpha_{DCN}(y_t^{PC1} - y_t^{PC2}) \tag{6}$$

where $\alpha_{DCN} \in [0, 1]$ represents the integration rate.

2.4 2.4. Kinematic Readout

The filtered DCN activity drives the kinematic drift rate:

$$v_t = \beta_v \cdot D_t \tag{7}$$

3 3. Empirical Benchmarks and Statistical Significance

Evaluated via 70:30 cross-validation on $N = 30$ human participants, the Unified Temporal Integration model achieved the lowest Test NLL in the history of the project.

Model Structure	Aggregate Test NLL	p-value vs M1
M1: Win-Stay/Lose-Shift (WSLS)	1448.57	-
M2: Random Walk Cognitive Filter	1623.98	-
ECCM: Unified Temporal Integration	**1433.15**	**0.00066**

Table 1: Final 3-way benchmark demonstrating strict statistical superiority.